AI Programming Foundations Project.

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Overview

This project demonstrates a complete data analysis workflow using Python and Jupyter Notebook to explore Airbnb listing data in New York City. The goal was to clean, analyze, and visualize the dataset to uncover patterns in pricing, neighborhoods, and reviews. The project follows best practices in data science, including reproducibility, structured workflows, and clear interpretation of results.

Dataset Description

The dataset used in this project is the **New York City Airbnb Open Data**, available on Kaggle: https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data

This dataset contains information about Airbnb listings in New York City from 2019, including variables such as price, neighborhood, room type, number of reviews, and availability. It includes approximately 48,000 rows and multiple columns describing listing characteristics and host information.

Key variables analyzed in this project include:

- `price` (listing cost)
- `neighbourhood_group` (borough)
- `room_type`
- `number_of_reviews`

These variables were selected to better understand pricing patterns and how they vary across different locations and listing types.

Workflow Description

1. Data Ingestion

The dataset was loaded into a Pandas Data Frame using Python. Initial inspection was performed to understand structure, missing values, and data types.

2. Data Cleaning

Cleaning steps included:

- Handling missing values in review-related columns
- Converting data types where necessary
- Removing or correcting inconsistent entries

Data cleaning is essential because poor data quality can negatively impact analysis and conclusions (Pimentel et al., 2019).

3. Exploratory Data Analysis (EDA)

EDA included:

- Summary statistics
- Correlation analysis
- Distribution checks for price and reviews

This step helped identify trends and outliers before visualization.

4. Visualizations

Three key visualizations were created:

- Price distribution histogram
- Neighborhood comparison (price by borough)
- Relationship between price and number of reviews

Visualizations were used to communicate patterns clearly and effectively.

5. Summary

The workflow concluded with interpretation of findings and insights about Airbnb pricing behavior across NYC.

Key Decisions and Assumptions

Several important decisions were made during the analysis:

Cleaning Choices

Missing values in review-related columns were handled to ensure consistency. This aligns with best practices, as missing data can bias results if not addressed properly (Van Buuren, 2018).

EDA Focus

The analysis focused on price, neighborhood, and reviews because these are key drivers of Airbnb market dynamics.

Visualization Design

- The histogram was used to show price distribution and identify skewness.
- The neighborhood comparison plot highlights geographic differences.
- The scatter plot shows relationships between engagement (reviews) and pricing.

These choices follow visualization best practices, where each chart is designed to answer a specific question.

Results and Interpretation

The analysis revealed several important patterns:

- **Price Distribution (Figure 1):** Prices are highly right-skewed, with most listings at lower prices and a few extreme high values.
- **Neighborhood Differences (Figure 2):** Manhattan listings tend to have significantly higher prices compared to other boroughs.
- **Price vs Reviews (Figure 3):** Listings with more reviews often have moderate pricing, suggesting competitive pricing strategies.

These findings highlight how location and demand influence Airbnb pricing.

Responsible Practice

Data cleaning and preprocessing decisions can introduce bias. For example:

- Removing missing values may exclude certain types of listings
- Outliers in price may distort interpretations if not handled carefully

To reduce bias:

- Cleaning decisions were kept minimal and transparent
- Extreme values were analyzed rather than blindly removed

Future improvements could include more robust outlier handling and bias analysis.

Reproducibility

This project is fully reproducible. To rerun the analysis:

1. Please clone the Git repository from the **master branch**, as it contains the production-ready codebase. The **development branch** is available for continuous improvements and future updates.
2. Install dependencies using `pip install requirements.txt`
3. Run the Jupyter notebook (`data_workflow.ipynb`)
4. Execute all cells in order

Using version control (Git) ensures that changes are tracked and reproducible. Reproducibility is a core principle of data science, allowing others to validate and build upon results (Pimentel et al., 2019).

References

Pimentel, J. F., Murta, L., Braganholo, V., & Freire, J. (2019).
Reproducible Data Science with Python: An Open Learning Resource.

Van Buuren, S. (2018).
Flexible Imputation of Missing Data. CRC Press.

Kaggle Dataset:
New York City Airbnb Open Data
https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data